

5-7 Nonrelativistic two-body collisions in 3 dimensions (Vol. I, Chs.10 and 11)

****7-1** A moving particle of mass M collides perfectly elastically with a stationary particle of mass $m < M$. Find the maximum possible angle through which the incident particle can be deflected.

****7-2** An object of mass m_1 , moving with a linear speed v in a laboratory system, collides with an object of mass m_2 which is at rest in the laboratory. After the collision, it is

observed that $(1 - \alpha^2)$ of the kinetic energy in the CM system was lost in the collision. What was the percentage loss of energy in the *laboratory* system?

****7-3** A proton with kinetic energy 1 MeV collides elastically with a stationary nucleus and is deflected through 90° . If the proton's energy is now 0.80 MeV, what was the mass of the target nucleus in units of the proton mass?

5-8 Forces (Vol. I, Ch. 12)

***8-1** Two masses, $m_1 = 4$ kg and $m_3 = 2$ kg, are connected with cords of negligible weight over essentially frictionless pulleys to a third mass, $m_2 = 2$ kg. The mass m_2 moves on a long table with a coefficient of friction

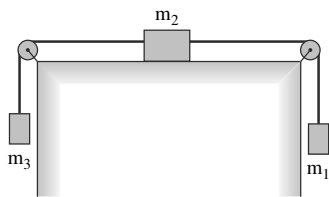


FIGURE 8-1

$\mu = 1/2$. What is the acceleration of mass m_1 after the system is released from rest?

****8-2** A 5 g bullet is fired horizontally into a 3 kg wooden block resting on a horizontal surface. The coefficient of sliding friction between the block and surface is 0.2. The bullet remains embedded in the block, which is observed to slide 25 cm along the surface. What was the velocity of the bullet?